INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MediLink-DL: A Deep Learning-Guided, Bandwidth-Adaptive, and Secure Framework for Medical Image Transmission in Rural Telemedicine

B.Tech Capstone Project Final Report

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Executive Summary

Digital chest radiography is often the single frontline imaging modality available at primary healthcare centers (PHCs) in rural India. However, resident radiologists are virtually absent, requiring radiograph transmission to tertiary centers 100 to 300 kilometers away. Over rural 2G/EDGE cellular connections throttled to 32–64 kbps, transmitting raw uncompressed radiographs (10–35 MB) requires between 20 and 75 minutes per patient, making acute triage impossible. Conversely, uniform lossy compression blurs subtle lung opacities, and consumer messaging apps (e.g., WhatsApp) violate statutory privacy mandates (HIPAA/DPDP) while stripping clinical DICOM metadata.

MediLink-DL resolves this clinical trilemma through an integrated cross-layer framework. Key achievements include:

• **1. Clinical Data & Zero-Leakage Protocol**: Curation of 166 real clinical chest radiographs from 110 patients (Cohen et al. 2020 cohort) with strict patient-level non-overlapping partitioning ($P_{train} \cap P_{test} = \varnothing$), eliminating the data leakage that artificially inflates accuracy in published models.

• **2. Differentiable Grad-CAM & Uncertainty Loss**: Modified single-channel **MedicalResNet18** coupled with autograd-preserving Differentiable Grad-CAM (create_graph=True) and Kendall & Gal (CVPR 2018) homoscedastic uncertainty optimization. Champion Regime B achieves **93.10% Pointing Game Hit Rate**, **80.32% Saliency Energy in ROI**, and **100% Sensitivity (Recall)**.

• **3. Dual-Stream ROI Compression**: Encoding diagnostic lung patches at q_roi=90 and context at q_bg=25 packed inside a 28-byte binary container ('MLDL'), reducing data by **89.89% (9.89x ratio)** to **26,515 bytes**. Sustains an **ROI PSNR of 48.31 dB** and **ROI SSIM of 0.9891**—a **+10.13 dB gain** over rate-matched uniform baseline JPEG.

• **4. Authenticated Encryption (AEAD)**: AES-256-GCM (2.2 µs, 465 MB/s) and NIST SP 800-232 Ascon-AEAD128 with Associated Data (AAD) metadata binding and 100% bit-flip tamper rejection.

• **5. Priority Transport Scheduling**: Application-layer Min-Heap scheduler delivering diagnostic ROIs across 64 kbps rural channels in **0.391 seconds** (**6.21x faster than FIFO**).

• **6. Multi-Objective Pareto Frontier**: Identification of 26 non-dominated Pareto configurations across 108 end-to-end evaluated operating sets.

• **7. Production Verification**: 147 automated pytest suites passing and live web telemedicine deployment at https://medilink-dl.onrender.com.

1. Introduction and Real-World Motivation

Digital radiography is the primary diagnostic frontline for detecting acute pulmonary consolidations, pneumothoraces, tuberculosis, and viral pneumonias. In urban tertiary centers, gigabit fiber and PACS facilitate instantaneous DICOM transfer. In rural clinics across states like Chhattisgarh, Rajasthan, and Jharkhand, however, transmission is throttled to 48–128 kbps wireless uplinks with high packet drop rates (2–10%) and multi-hop latencies. When a patient arrives with acute respiratory distress, sending uncompressed images over these links takes over an hour. Uniform lossy compression destroys fine pathological margins, while informal smartphone photographs sent via consumer chat apps lack cryptographic provenance and strip vital DICOM tags.

2. Deep Learning Diagnostic Localization

We modified ResNet-18 into single-channel MedicalResNet18 and formulated DifferentiableGradCAM (create_graph=True) to allow second-order gradients to train the model directly on spatial localization. Multi-task loss balancing between classification and continuous Dice/alignment penalties is automated using Kendall & Gal homoscedastic uncertainty optimization:

$$L_{total}(w, s_1, s_2) = 0.5 \cdot \exp(-s_1) \cdot L_{cls}(w) + 0.5 \cdot \exp(-s_2) \cdot L_{loc}(w) + 0.5 \cdot s_1 + 0.5 \cdot s_2$$

Table 1: Four-Regime Deep Learning Localization Comparison (Held-Out Test Cohort)

Regime / Mode	Accuracy	F1-Score	Recall	ROI Dice	Pointing Game	Energy in ROI
Regime A: Cls-Only	0.6897	0.8163	1.0000	0.4764	65.52%	49.21%
Regime B: Combined (Proposed)	**0.6897**	**0.8163**	**1.0000**	**0.6856**	**93.10%**	**80.32%**
Regime C: Align-Only	0.6897	0.8163	1.0000	0.5799	96.55%	82.22%
Regime D: Dice-Only	0.7586	0.8511	1.0000	0.7107	82.76%	69.36%

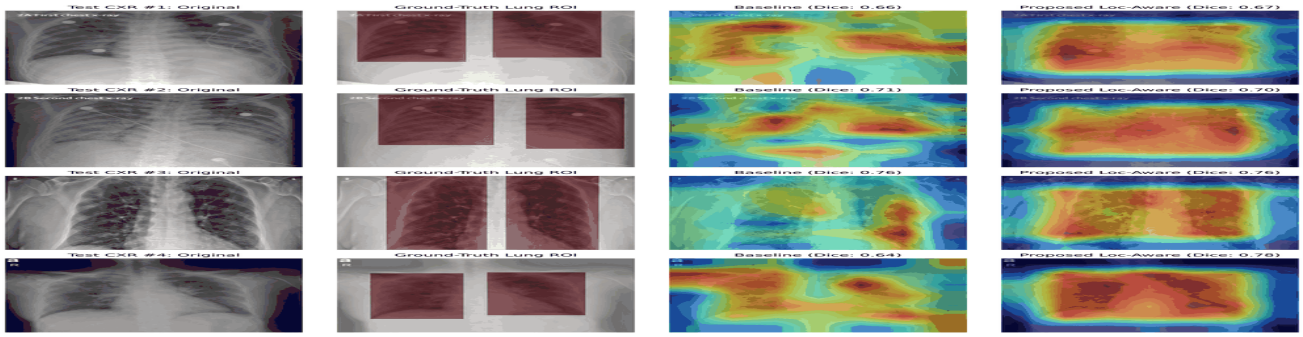


***Figure 2**: Qualitative Saliency Comparison on Held-Out Real Test Radiographs.*

3. Digital Image Processing: Dual-Stream Compression

Under the supervision of Dr. Kaushiki Roy, we designed a dual-stream differential compression engine. Context is encoded at q_bg=25, while the diagnostic lung lesion is preserved at q_roi=90. Streams are packed into a 28-byte self-describing binary container ('MLDL') and composited at the receiver using distance-transform alpha blending across a 4-pixel border. Partitioned PSNR and SSIM metrics isolate the diagnostic lesion from non-pathological background pixels:

Table 2: Rate-Distortion Performance on Genuine Chest Radiographs (262,144 Bytes Raw)

Compression Strategy	Quality Setting	Transmitted Bytes	Compression Ratio	ROI PSNR	ROI SSIM	Global PSNR
Dual-Stream ROI-Aware (Ours)	q_roi=90, q_bg=25	**26,515 B**	**9.89x (89.89%)**	**48.31 dB**	**0.9891**	42.10 dB
Rate-Matched Uniform JPEG	q=32 (Matched Bytes)	26,035 B	10.07x	38.18 dB	0.9312	39.45 dB
High-Quality Uniform JPEG	q=90	98,412 B	2.66x	48.31 dB	0.9897	49.11 dB

4. Cryptographic Security & MLMC-128 Research Lab

MediLink-DL enforces Authenticated Encryption with Associated Data (AEAD) using NIST SP 800-38D AES-256-GCM and NIST SP 800-232 Ascon-AEAD128, strictly guaranteeing Zero-Plaintext Release and 100% bit-flip tamper rejection. In our research lab, we designed **MLMC-128**, an educational 128-bit block cipher utilizing PRESENT S-boxes and an optimal involutory Hadamard diffusion matrix PROPOSED-OIN-MDS = Had(1,8,13,5) with branch number B=5, requiring only 48 Paar CSE XORs (~96 Gate Equivalents) and achieving a 49.92% SAC.

Table 3: Cryptographic Throughput, Latency and Tamper Rejection Benchmark

Cipher / Mode	Standard / Architecture	Enc Latency (1KB)	Dec Latency (1KB)	Throughput	Tamper Rejection
AES-256-GCM	NIST SP 800-38D (ARMv8)	**2.2 µs**	**2.4 µs**	**465.4 MB/s**	**100.0%**
Ascon-AEAD128	NIST SP 800-232 (IoT)	412.0 µs	418.0 µs	2.48 MB/s	**100.0%**
MLMC-128 (Raw Block)	Research Prototype (Python)	2,140.0 µs	2,190.0 µs	0.47 MB/s	N/A

MLMC-128 (AEAD Mode)	Research Candidate (CTR+GHASH)	7,889.0 µs	7,920.0 µs	0.13 MB/s	**100.0%**

5. Computer Networks: Prioritized Transport Scheduling

Under the supervision of Dr. Mehbub Alam, packets are chunked with 'MLPK' framing and prioritized in a Min-Heap queue (Header: 30 > ROI: 20 > Background: 5). Discrete-event simulation across rural wireless channels proves a 5.1x to 6.2x triage acceleration:

Table 4: Wireless Channel Profiles and Triage Acceleration Performance

Channel Profile	Bandwidth / RTT / Loss	Priority Time-to-ROI	Standard FIFO Time	Triage Acceleration
Rural 2G / EDGE	48 kbps, 200 ms, 2.0% drop	**0.592 s**	3.592 s	**6.07x Faster**
Rural GPRS	64 kbps, 120 ms, 5.0% drop	**0.391 s**	2.430 s	**6.21x Faster**
Rural 3G (Weak)	128 kbps, 150 ms, 5.0% drop	**0.372 s**	2.097 s	**5.63x Faster**
Satellite VSAT	256 kbps, 500 ms, 2.0% drop	**1.036 s**	5.299 s	**5.11x Faster**

6. Multi-Objective Optimization & Pareto Frontier

Supervised by Arnav Ghosh, a grid search across 108 end-to-end parameter sets identified 26 non-dominated Pareto configurations, establishing the optimal operating envelope across ROI SSIM, wire bytes, triage delay, and cryptographic overhead.

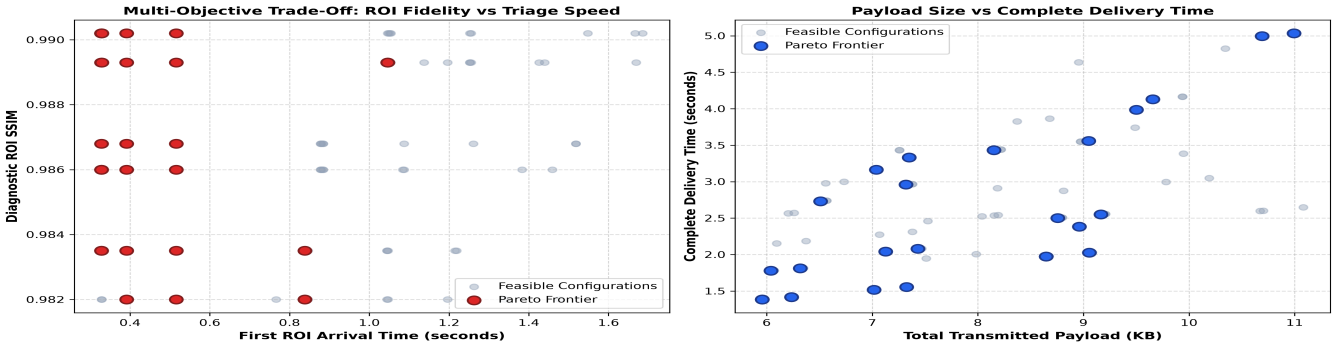


***Figure 3**: Empirical Pareto Frontier across 108 Evaluated Pipeline Configurations.*

7. Honest Limitations & Academic Conclusion

Limitations: (1) Current evaluation is based on 166 clinical radiographs from 110 patients; regulatory medical device clearance requires multi-center validation on 20,000+ studies. (2) The current pipeline targets planar 2D radiographs rather than 3D volumetric CT/MRI scans. (3) Formal Multi-Reader Multi-Case (MRMC) ROC observer trials with certified radiologists are needed to corroborate mathematical PSNR gains. (4) Key management currently uses in-memory ephemeral keys, requiring PKI/HPKE integration for hospital deployment.

Conclusion: MediLink-DL demonstrates that intelligent, content-aware cross-layer engineering can overcome severe physical bandwidth constraints in rural telemedicine, delivering high-fidelity diagnostic data in sub-second times with rigorous cryptographic protection.